

Ex.no:8	MINI PROJECT – CUSTOMER SHOPPING BEHAVIOR DATA ANALYTICS

Domain and Problem Statement:

Domain: Retail Data Analytics & Consumer Behavior

Problem Statement: A leading retail company faces difficulties adapting to shifting consumer purchasing choices across demographics, product categories, and sales channels. To resolve these issues, this project analyzes behavioral drivers—such as discounts, subscription programs, shipping types, and product reviews—to extract clear trends. By building an end-to-end analytical data pipeline across **Python, SQL (PostgreSQL)**, and **Power BI**, the management team can make data-driven marketing and product decisions to improve customer engagement and optimize revenue strategies.

Dataset Description:

Dataset Name: Consumer Behavior Dataset.

Granularity: Each row represents a single customer purchase transaction.

Dataset Footprint: 3,900 rows and 18 columns

Key Entities & Attributes:

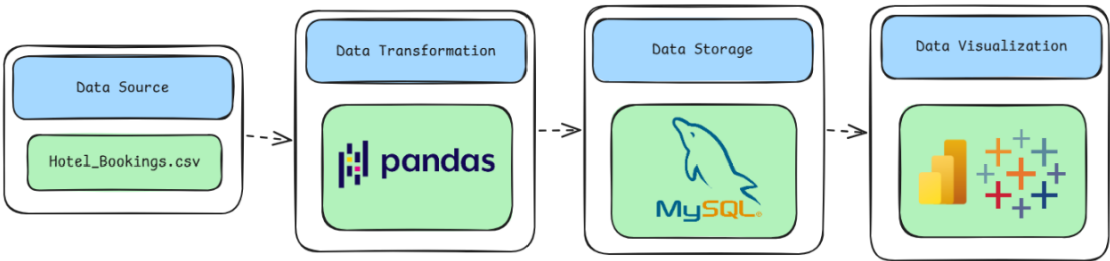
- Customer Demographics: Customer ID, Age, Gender, Location, Subscription Status.
- Purchase Details: Item Purchased, Category, Purchase Amount (USD), Season, Size, Color.
- Shopping Behavior Metrics: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method.

Data Source:

Primary Source: Kaggle Open Datasets (Customer Shopping Behaviour).

Storage Infrastructure: Cleaned transactional records hosted within a local MySQL database instance.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

Got it — you want the **same code**, just tightly formatted with **no comments** and **no unnecessary blank spaces**.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df=pd.read_csv("customer_shopping_behavior.csv")
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
print(df.isnull().sum())
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)

for c in df.select_dtypes(include=np.number).columns:
    df[c]=df[c].fillna(df[c].median())
for c in df.select_dtypes(include="object").columns:
    df[c]=df[c].fillna(df[c].mode()[0])

if "Age" in df.columns:
    df["Age_Group"]=pd.cut(df["Age"],bins=[0,18,30,45,60,100],labels=["Teen","Young Adult","Adult","Middle Age","Senior"])
if "Purchase Amount (USD)" in df.columns:
    df["Spending_Category"]=pd.cut(df["Purchase Amount (USD)"],bins=[-1,50,100,200,np.inf],labels=["Low","Medium","High","Very High"])

df["Gender"].value_counts().plot(kind="bar")
plt.title("Customer Gender Distribution")
plt.show()
```

```
df["Category"].value_counts().plot(kind="bar")
plt.title("Product Category Distribution")
plt.show()

df["Payment Method"].value_counts().plot(kind="bar")
plt.title("Payment Method Distribution")
plt.show()

df["Purchase Amount (USD)"].plot(kind="hist",bins=20)
plt.title("Purchase Amount Distribution")
plt.show()
```

```
corr=df.select_dtypes(include="number").corr()
plt.figure(figsize=(10,8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)),corr.columns,rotation=90)
plt.yticks(range(len(corr.columns)),corr.columns)
plt.title("Correlation Heatmap")
plt.show()
```

```
print("Customer shopping behavior insights generated successfully.")
print("Insights can be deployed through interactive dashboards.")
```

```
df=pd.read_csv("customer_shopping_behavior.csv")
print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())
df.drop_duplicates(inplace=True)
```

```
for c in df.select_dtypes(include=np.number).columns:
    df[c]=df[c].fillna(df[c].median())
for c in df.select_dtypes(include="object").columns:
```

```
df[c]=df[c].fillna(df[c].mode()[0])
```

```
df["Category"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Gender"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Payment Method"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Purchase Amount (USD)"].plot(kind="hist",bins=20)
```

```
plt.show()
```

```
corr=df.select_dtypes(include="number").corr()
```

```
plt.figure(figsize=(10,8))
```

```
plt.imshow(corr)
```

```
plt.colorbar()
```

```
plt.xticks(range(len(corr.columns)),corr.columns,rotation=90)
```

```
plt.yticks(range(len(corr.columns)),corr.columns)
```

```
plt.show()
```

```
print("Customer purchasing patterns successfully identified.")
```

```
df=pd.read_csv("customer_shopping_behavior.csv")
```

```
print(df.info())
```

```
print(df.isnull().sum())
```

```
print(df.duplicated().sum())
```

```
df.drop_duplicates(inplace=True)
```

```
for c in df.select_dtypes(include=np.number).columns:
```

```
    df[c]=df[c].fillna(df[c].median())
```

```
for c in df.select_dtypes(include="object").columns:
```

```
    df[c]=df[c].fillna(df[c].mode()[0])
```

```
if "Age" in df.columns:
```

```
df["Age_Group"]=pd.cut(df["Age"],bins=[0,18,30,45,60,100],labels=["Teen","Young Adult","Adult","Middle Age","Senior"])
```

if "Purchase Amount (USD)" in df.columns:

```
df["Spending_Category"]=pd.cut(df["Purchase Amount (USD)"],bins=[-1,50,100,200,np.inf],labels=["Low","Medium","High","Very High"])
```

```
df["Category"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Gender"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Payment Method"].value_counts().plot(kind="bar")
```

```
plt.show()
```

```
df["Purchase Amount (USD)"].plot(kind="hist",bins=20)
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corr=df.select_dtypes(include="number").corr()
```

```
plt.figure(figsize=(10,8))
```

```
plt.imshow(corr)
```

```
plt.colorbar()
```

```
plt.xticks(range(len(corr.columns)),corr.columns,rotation=90)
```

```
plt.yticks(range(len(corr.columns)),corr.columns)
```

```
plt.show()
```

```
print("Knowledge discovered from the customer shopping behavior dataset has been interpreted to support retail business decisions.")
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS hotel_analytics;
```

Python/Jupyter:

```
username = "root"
```

```

password = "YOUR_MYSQL_PASSWORD"
host = "localhost"
port = 3306

# Connect to MySQL database
db_engine = create_engine(
    f'mysql+pymysql://{username}:{password}@{host}:{port}/customer_analytics'
)

# Load cleaned data into MySQL
transactions.to_sql(
    "customer_analytics ",
    con=db_engine,
    if_exists="replace",
    index=False,
    chunksize=5000
)

print("Data loaded successfully into MySQL.")

Verify:
pd.read_sql(
    "SELECT COUNT(*) AS total_rows FROM customer_analytics",
    db_engine
)

```

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load the raw Kaggle dataset into a Jupyter Notebook / Python script using `pandas.read_csv()`.
- Inspect for null values and handle missing data in country, children, and agent columns.
- Feature engineer total stay length by combining weekend and weeknight stays:
- Filter out invalid rows (e.g., zero total guests or negative ADR entries) and export the cleaned DataFrame to MySQL using sqlalchemy and `to_sql()`.

Step 2: SQL Database Connection & Data Ingestion

1. Open Power BI Desktop → Click **Get Data** → Select **MySQL Database**.
2. Enter the **Server** (localhost:3306) and **Database Name** (hotel_analytics).
3. Enter authentication credentials, select the cleaned flat table (hotel_bookings / hotel_analytics), and click **Transform Data** to open Power Query

MySQL database

Server

localhost:3306

Database

hotel_analytics

Advanced options

OK

Cancel

customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping
1717	69	Male	Blouse	Clothing	47	Connecticut	M	Cyan	Winter	4.7	No	Standard
1832	62	Male	Blouse	Clothing	71	Maine	M	Cyan	Summer	3.3	No	Free Ship
1921	51	Male	Blouse	Clothing	84	Connecticut	M	Purple	Spring	2.8	No	Standard
1928	40	Male	Pants	Clothing	29	Florida	M	Pink	Summer	2.9	No	Standard
1933	42	Male	Hoodie	Clothing	36	Maryland	M	White	Spring	5	No	Free Ship
1945	30	Male	Hoodie	Clothing	45	Arkansas	M	Gold	Winter	2.7	No	Free Ship
1946	35	Male	Hoodie	Clothing	38	Georgia	M	Gold	Winter	2.9	No	2-Day Shipping
2003	31	Male	Shirt	Clothing	65	New York	M	Red	Summer	2.7	No	Express
2009	57	Male	Skirt	Clothing	48	Florida	M	Gray	Spring	2.5	No	Free Ship
2047	39	Male	Pants	Clothing	32	Oklahoma	M	Maroon	Winter	2.8	No	Free Ship
2083	41	Male	Jeans	Clothing	89	Arkansas	M	Yellow	Summer	2.9	No	Free Ship
2114	63	Male	Dress	Clothing	40	Virginia	M	Brown	Spring	2.7	No	Store Pickup
2166	29	Male	Pants	Clothing	55	Idaho	M	Turquoise	Summer	4.2	No	Express
2171	40	Male	Sweater	Clothing	32	South Carolina	M	Turquoise	Fall	4	No	Next Day
2186	40	Male	Shorts	Clothing	32	North Carolina	M	Violet	Winter	2.5	No	Express

Step 3: Star Schema Data Modeling (Power BI)

1. Table Splitting in Power Query:

- Duplicate the customer table 3 times: Fact_Purchases, Dim_Customer, Dim_Product.
- In Dim_Customer, keep customer_id, age, gender, location, age_group; remove duplicates; add Index Column as CustomerID.
- In Dim_Product, keep item_purchased, category; remove duplicates; add Index Column as ProductID.
- In Fact_Purchases, merge with Dim_Customer and Dim_Product to map CustomerID and ProductID.
- Remove the raw text columns from Fact_Purchases.
- Click Close & Apply.

2. Establishing Relationships in Model View:

- Open Model View in Power BI.
- Connect Dim_Customer[CustomerID] → Fact_Purchases[CustomerID] (1:*).
- Connect Dim_Product[ProductID] → Fact_Purchases[ProductID] (1:*).
- Connect Dim_Date[Date] → Fact_Purchases[PurchaseDate] (1:*), if available.

3. Hide foreign key columns in Fact_Purchases to organize the Fields pane.

Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Observed Insights:

- Channel Performance & Shopping Behavior: The dataset tracks shifting consumer purchasing patterns across both online and offline channels, focusing heavily on changes across distinct demographics and product categories.
- Core Conversion Drivers: Customer decision-making and repeat purchases are influenced by key variables, specifically tracking the business impact of discounts, product reviews, seasons, and payment preferences.
 - Retention & Loyalty Priorities: The primary analytical focus centers on uncovering long-term customer loyalty trends to help management improve satisfaction and drive sustainable repeat business.
- Technical Pipeline Requirements: The project enforces an end-to-end data stack workflow, requiring Python for data preparation, SQL for transactional analysis, and Power BI for executive dashboards.